

Stephen Sun

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EXPERIENCE

Microsoft AI

Mar 2025 - Present

Software Engineer II - Bing Maps Data Infra team

Redmond, WA

- Led migration of Bing Maps data pipelines from proprietary internal tooling to open-source frameworks – Apache Spark, Iceberg, and Airflow, improving pipeline performance by 50% and ensuring long term support.
- Built and owned Scala Spark export jobs that processed world-scale geospatial data, powering maps rendering, search relevance, and grounding data workflows.
- Redesigned team's primary geospatial data store to store supplementary attributes such as business reviews.
- Improved pipeline reliability through introducing data validation checks and gating mechanisms.

ServiceNow

Jul 2022 - Mar 2025

Software Engineer - Big Data team

Santa Clara, CA

- Built and managed production Java Spark pipelines for querying, processing and visualizing ServiceNow customer data.
- Developed framework to parse raw log files, then correlate them with alerts and anomalies. Support and Site Reliability teams' Mean Time to Recover was improved from >1h to 1min per engineer, and >1000 company hours are saved yearly.
- Deployed key ServiceNow global instance availability data pipeline with minute-level data and daily and monthly aggregations. Data is used by 200+ users through real-time visualizations and aggregated reports.
- Onboarded critical pipeline to ingest user experience data from >20,000 customers for usage and performance analytics.

Brown University

Sep 2021 - Dec 2021

Teaching Assistant - CSCI 1600: Real-Time and Embedded Software

Providence, RI

- Developed course material for TA-led labs with Prof Milda Zizyte for the inaugural run of the class, with 35 students.
- Ran labs using Arduino MKR1000, mentored student groups for a semester long design project and graded homework.

ServiceNow

May 2021 - Aug 2021

Software Engineer Intern - Big Data team

Santa Clara, CA

- Primary developer for a logging framework for BigData team's MariaDB database to track all CRUD operations. Framework is still regularly used today for tracking changes and debugging.
- Created Jenkins job to regularly check 6000+ metrics for discrepancies in metadata, then alert engineers for resolution.
- Participated in ServiceNow's internal hackathon and was part of the top 8 finalists out of over 50 teams.

EDUCATION

Brown University

Sep 2018 - May 2022

Bachelor of Science: ScB - Computer Science and AB - Engineering

Providence, RI

- **Coursework:** CS - Operating Systems, Machine Learning, Data Science, Programming Languages, Distributed Systems
Engineering - Design of Robotic Systems, Digital Electronics System Design, Design of Computing Systems
- **Leadership:** President - Brown University Merlions (Singapore Student Club), Co-Founder/VP - Brown Flying Club

PROJECTS

Undergraduate Teaching and Research Award - CubeSat Robot Arm

- Built proof-of-concept robotic arm for use on nanosatellites under Brown School of Engineering's Prof Rick Fleeter. Arm was designed in SolidWorks, control software was coded in MATLAB and Arduino using Inverse Kinematics concepts.
- Prototype arm and software serves as the foundation for the Brown Space Engineering student club's PVDX CubeSat, and project was featured on Brown University's news portal.

Independent Study Course Project - Indoor Robot Navigation with 3D LiDAR

- Programmed TurtleBot 2 in C++ to autonomously navigate through an indoor environment using Simultaneous Location and Mapping (SLAM) and people detection algorithms, while equipped with a Velodyne VLP-16 LiDAR.

Weenix OS - Barebones Unix Operating System

- Constructed Unix-like operating system in C that is able to make system calls, run system commands and user code.

SKILLS

Languages: Scala, Java, Python, SQL, Bash, C, C++

Data Processing & Analytics: Apache Spark, Airflow, Kafka, Parquet, Kudu, Impala, Grafana, InfluxDB

Storage & Data Formats: Apache Iceberg, Parquet, HDFS, HBase, Kudu

Cloud Platforms: Azure Data Factory, Azure Synapse, Google Cloud Platform, Cloudera Data Platform